

CSG3101 Applied Project

Project Report For:

Intelligent Spatial Management System

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Abstract

This project presents the development of an intelligent room booking system tailored for Edith Cowan University (ECU) Sri Lanka, addressing inefficiencies in resource management and enhancing user experience. The primary objective was to create a responsive web application to streamline room bookings, optimize resource allocation, and reduce conflicts through real-time updates, advanced notifications, and AI-powered assistance.

The system was developed using **Agile with Scrum** methodology, ensuring iterative progress, continuous feedback, and adaptability to changing requirements. The development cycle included multiple sprints focusing on implementing core functionalities such as user and role management, room availability tracking, penalty systems, and advanced analytics. An NLP-powered chatbot, integrated with Wit.ai, was developed to assist users in navigating the application and managing bookings effortlessly.

The key results include:

1. **User Management:** A secure system for user registration, role-based access control, and account management.
2. **Room Management:** Real-time availability updates, room usage logging, and admin-controlled approvals.
3. **Penalty System:** Automated tracking and reporting of penalties for no-shows, late cancellations, and overstays.
4. **Notifications:** Alerts for booking updates, proactive overstay warnings, and energy efficiency reminders.
5. **Analytics:** Comprehensive insights into resource utilization trends, predictive analytics using Prophet, and exportable reports.
6. **Chatbot Assistance:** An AI-driven chatbot enabling dynamic interaction and personalized support for room searches and bookings.

The project successfully met the majority of the proposed goals, with deferred functionalities like waitlist management and customizable dashboards noted for future enhancements. Supervisor-approved changes, including soft deletion for room records and CSV exportable reports, ensured the system's robustness and administrative utility.

In conclusion, the intelligent room booking system provides a scalable, user-friendly solution to resource management challenges at ECU Sri Lanka. It demonstrates how integrating cutting-edge technologies such as AI, predictive analytics, and Scrum-based iterative development can significantly improve operational efficiency and user satisfaction. Future enhancements, such as waitlist management and mobile app integration, are planned to further enhance the system's impact and adaptability.

Introduction

Problem Statement

At Edith Cowan University (ECU) Sri Lanka, managing shared spaces such as lecture rooms, study areas, and seminar halls is a growing challenge. The absence of a centralized booking system leads to inefficiencies, including scheduling conflicts, underutilized spaces, and user frustration. With the increasing demand for campus resources and a diverse range of users—students, lecturers, and administrative staff—these issues have a significant impact on day-to-day operations.

For instance, overlapping bookings and the lack of visibility into room availability often cause delays and disrupt academic schedules. Furthermore, the manual processes for managing room usage and approving bookings place an unnecessary administrative burden on staff. In a rapidly expanding campus environment, it is critical to address these challenges with a scalable, efficient, and user-friendly solution.

Objectives

The primary objective of this project is to design and implement a responsive web application tailored to the needs of ECU Sri Lanka. The system aims to improve campus operations by optimizing room bookings and addressing existing inefficiencies through advanced technology. The specific goals include:

1. **Real-Time Room Availability:**
 - Provide live updates on room status, enabling users to book and manage rooms effectively.
 - Ensure role-based access so that students, lecturers, and admins can interact with the system according to their specific needs.
2. **Automated Penalty Mechanism:**
 - Implement a penalty system to manage overstays, late cancellations, and no-shows, ensuring accountability among users.
 - Provide detailed tracking and reporting of penalties for better administrative oversight.
3. **User Notifications and Alerts:**
 - Deliver timely notifications for booking confirmations, reminders, and penalty updates to users.
 - Improve resource efficiency with proactive alerts, such as energy reminders for turning off room resources like air conditioning or lights.
4. **Comprehensive Analytics and Reporting:**
 - Offer insights into room utilization trends, peak periods, and user behaviors through an analytics dashboard.
 - Enable the export of resource utilization data for admin review and strategic planning.
5. **AI-Driven Chatbot Assistance:**
 - Develop an AI-powered chatbot to assist users in navigating the system, checking room availability, and managing bookings.
 - Provide personalized suggestions for room allocation and troubleshoot common issues efficiently.

Importance of the Project

This project is crucial for ECU Sri Lanka to modernize its resource management and meet the growing demands of an expanding campus. The implementation of an intelligent room booking system will offer significant benefits in terms of:

1. Operational Efficiency:

- Automating booking approvals, availability tracking, and penalty management reduces administrative workload and eliminates manual errors.
- Real-time updates streamline resource allocation, ensuring smoother academic and extracurricular activities.

2. Enhanced User Experience:

- Role-based functionalities cater to the specific needs of students, lecturers, and admins, improving user satisfaction.
- Features like the AI-powered chatbot simplify the booking process and resolve queries quickly, reducing reliance on administrative support.

3. Data-Driven Decision Making:

- Advanced analytics enable ECU Sri Lanka to identify trends in room usage, optimize resource planning, and forecast future needs.
- Exportable reports allow administrators to share insights with stakeholders and make informed decisions about campus resource allocation.

4. Scalability and Future Growth:

- The modular architecture ensures the system can scale to accommodate additional users, buildings, and campuses as ECU Sri Lanka grows.
- Future enhancements, such as energy efficiency tracking and mobile app integration, will ensure the system remains relevant and impactful.

By addressing the unique challenges faced by ECU Sri Lanka, this project aims to deliver a comprehensive solution that enhances campus operations and aligns with the university's vision of excellence in resource management.

Executive summary

The project geared toward developing the Intelligent Spatial Management System which presented a user-friendly web application for ECU Sri Lanka's room booking administration purposes. The system enables live viewing of availability and equipped with automated financial penalties and artificial intelligence-based chat assistance for user support. The system achieved its objectives by improving operational efficiency alongside administrative burden reduction and better user satisfaction. Most proposed features of the project were successfully implemented using Agile Scrum methodology yet waitlist management capabilities remained scheduled for upcoming builds.

Adherence to Proposal

Direct Mapping of Proposal Goals to Executed Components

The project's execution adheres closely to the original proposal, with most functionalities fully implemented. Below is a detailed table mapping the proposal goals to the implemented components, categorized by their priority (Must Have, Could Have, Nice to Have).

Feature Category	Proposed Functionality	Priority	Implementation Status	Description
User Management	User Registration	Must Have	Fully Implemented	Register_user API allows user registration with validation for roles and duplicates.
	User Authentication	Must Have	Fully Implemented	login API provides JWT-based authentication for users.
	Role-Based Access Control	Must Have	Fully Implemented	Enforces role-specific permissions (student, staff, admin).
	Profile Management	Could Have	Fully Implemented	profile API enables retrieval and update of user details.
	Account Deactivation	Nice to Have	Fully Implemented	deactivate API allows admins to deactivate user accounts.
Room Management	Real-Time Room Availability	Must Have	Fully Implemented	room_availability API checks availability for specific time slots.
	Room Registration	Must Have	Fully Implemented	Admins can create, filter, soft delete, and approve rooms using RoomViewSet.
	Start and End Usage Options	Must Have	Fully Implemented	start_usage and end_usage APIs log room usage.
	Waitlist Management	Nice to Have	Not Implemented	No functionality exists for waitlist management or notifications.
Penalty System	Overstay Detection	Must Have	Partially Implemented	Penalties are applied for no-shows and late cancellations but not explicitly for overstays.
	Penalty Imposition	Must Have	Partially Implemented	Penalty tracking exists but does not reduce future booking durations dynamically.
	Penalty Tracking	Must Have	Fully Implemented	Penalties are stored and tracked via PenaltyViewSet.
	Repeat Offense Management	Could Have	Partially Implemented	Repeat offender data is logged, but no escalation mechanisms are implemented.
Notifications & Alerts	Booking Reminders	Must Have	Not Implemented	No functionality exists for booking reminders.
	Penalty Notifications	Must Have	Partially Implemented	General notifications are implemented for penalties but lack specific penalty-related notifications.

	Waitlist Notifications	Must Have	Not Implemented	Notifications for waitlist availability are not included.
	Proactive Overstay Alerts	Should Have	Fully Implemented	Alerts notify users of potential overstays using predictive algorithms.
	Energy Efficiency Reminders	Should Have	Fully Implemented	Notifications prompt users to check room resources like AC and lights before leaving.
	Custom Notification Preferences	Could Have	Not Implemented	No functionality exists for configuring notification preferences.
Analytics & Reporting	Resource Analytics	Must Have	Fully Implemented	APIs provide analytics data, including room usage, trends, and utilization.
	Feedback Analysis	Must Have	Fully Implemented	APIs collect and analyze user feedback for quality improvements.
	Exportable Reports	Must Have	Fully Implemented	CSV export APIs for analytics and penalties are implemented.
	Trend Prediction	Should Have	Fully Implemented	Prophet-based forecasting predicts future room usage trends.
	Real-Time Reporting	Could Have	Partially Implemented	Real-time metrics for active users and utilization rates are provided but lack dashboard integration.
	Customizable Dashboard	Nice to Have	Not Implemented	No functionality exists for customizable admin dashboards.
AI & ML Chatbot	User Assistance	Must Have	Fully Implemented	NLP-powered chatbot assists users with booking, availability checks, and cancellations.
	Dynamic Interaction	Must Have	Fully Implemented	Real-time intent recognition and responses using Wit.ai.
	Proactive Suggestions	Should Have	Partially Implemented	Chatbot provides suggestions for available rooms but lacks dynamic preference tracking.
	Error Resolution	Should Have	Fully Implemented	Chatbot resolves booking and availability issues effectively.
	Admin Escalation	Could Have	Fully Implemented	Chatbot escalates unresolved issues to administrators.

Table 1

Justifications for Variances

- **Fully Implemented:** Core functionalities, such as user and room management, analytics, and chatbot features, align with the proposal's goals.
- **Partially Implemented:** Features like penalty imposition and real-time reporting require further enhancements for complete functionality.
- **Not Implemented:** Features such as waitlist management and customizable dashboards were deprioritized to focus on core system stability.

Supervisor-Approved Changes

- **Room Management Enhancements:**
 - Added soft deletion for rooms to preserve historical data without impacting active listings.
- **Penalty Reporting:**
 - Introduced CSV export for penalties, improving admin workflows.
- **Energy Efficiency and Proactive Alerts:**
 - Supervisor approved adding these features to improve operational efficiency and user experience.
- **Deferred Features:**
 - Features such as waitlist management and notification preferences were excluded based on supervisor guidance to prioritize essential functionalities.

Stakeholder Interactions

Supervisory feedback became available through regular scheduled meetings. The supervisor recommended adopting soft deletion for room records while creating exportable penalty reports in the system. Tests involving real users resulted in system improvements as users provided feedback which shaped the responses of the chatbot for better usability.

Project Outputs and Deliverables

Completed Deliverables

Below is a list of all completed deliverables, their implementation details, and how they align with the project proposal:

Deliverable	Description	Implementation Details
User Registration	Allows users to register by providing email, username, password, and role.	Implemented via the <code>register_user</code> API in <code>users - views.py</code> , with validation for duplicates and role-specific restrictions.
User Authentication	Secure login functionality with JWT tokens for role validation.	Implemented via the <code>login</code> API in <code>users - views.py</code> , ensuring proper role-based authentication and session management.
Role-Based Permissions	Enforces access controls for different roles (student, staff, admin).	Managed across all APIs, restricting access to admin-specific features like room approval and penalty resolution.
Room Management APIs	Supports room creation, filtering, approval, and availability checks.	<code>RoomViewSet</code> provides APIs for real-time availability, soft deletion, and dynamic filtering by capacity and location.
Room Usage Logging	Logs start and end times for room usage.	Implemented via <code>start_usage</code> and <code>end_usage</code> actions in <code>rooms - views.py</code> .
Penalty Tracking	Tracks penalties for no-shows, late cancellations, and overstays.	Implemented via <code>PenaltyViewSet</code> in <code>penalties - views.py</code> , supporting user-specific queries and reporting.
Penalty Reporting	Generates CSV reports of penalties for admin review.	Implemented in the <code>download_report</code> action in <code>penalties - views.py</code> .

Notifications	Provides notifications for booking updates (approval, cancellation) and penalties.	Implemented via NotificationViewSet in notifications - views.py.
Analytics APIs	Aggregates room usage data and generates insights.	AnalyticsViewSet provides filtering and visualization-ready analytics data.
Exportable Reports	Enables admins to download analytics data as CSV files.	Implemented in ExportCSVView in analytics - views.py.
Room Usage Forecasting	Predicts future room usage trends using machine learning.	Implemented via Prophet-based forecasting in analytics - views.py.
Chatbot	NLP-powered chatbot assists users with room availability checks, bookings, and cancellations.	Implemented in chatbot_service.py, using Wit.ai for intent recognition and response generation.
Feedback Analysis	Collects and analyzes user feedback for quality improvement.	Implemented via APIs in analytics - views.py, enabling insights into user experiences.
Proactive Overstay Alerts	Alerts users about potential overstays using predictive algorithms.	Implemented in the penalty and analytics systems, ensuring efficient resource usage.
Energy Efficiency Reminders	Prompts users to turn off room resources like AC and lights before leaving.	Implemented as part of the notification system in notifications - views.py.
Admin Escalation (Chatbot)	Escalates unresolved issues to administrators for resolution.	Implemented in chatbot_service.py as part of error-handling and escalation workflows.

Table 2

Pending or Excluded Deliverables

Some features were not implemented or only partially completed due to resource constraints or scope prioritization.

Deliverable	Reason for Exclusion or Pending Status
Waitlist Management	Deferred due to time constraints; focus was placed on implementing core room management features.
Customizable Notification Preferences	User-configurable notification settings were excluded as they were not critical to the project goals.
Customizable Dashboards	Admin dashboards with configurable metrics and filters were excluded as they were considered non-essential.

Table 3

Reasons for Variances

1. Focus on Core Features:

- Resources were allocated to ensure the stability and robustness of key features like room management, penalties, and analytics.

2. Supervisor Recommendations:

- Features like penalty reporting, forecasting, and proactive alerts were prioritized based on supervisor feedback to improve operational efficiency.

3. Future Enhancements:

- Deferred features, such as waitlist management and customizable dashboards, are noted for future implementation in an expanded version of the system.

Quality Indicators

1. Evaluation of Output Quality

Situation: The project deliverables include various components like APIs, user interfaces, analytics, and chatbot integrations, all aimed at providing a seamless and efficient room booking system. The expectation was to develop a system that is functional, user-friendly, and scalable while adhering to real-world usability standards.

Action: Each component underwent rigorous development and validation processes:

- **APIs:** Functional testing ensured each endpoint worked as expected with correct inputs and error handling for invalid data.
- **Frontend:** Interfaces were designed with responsive layouts, ensuring usability across different devices (mobile, tablet, and desktop).
- **Analytics:** Predictive features and dynamic reporting were implemented and tested for accuracy and actionable insights.
- **Chatbot:** Integration with Wit.ai enabled natural language processing to simplify user interactions.
- **Error Handling:** Detailed feedback and error messages were added to maintain robustness.

Output: The system successfully meets its functional and non-functional requirements, producing consistent outputs that are usable, reliable, and intuitive. It has been validated through functional tests, manual user testing, and integration checks.

2. Generalizability of Outcomes

Situation: The system was designed to address the challenges of room management in educational institutions. A scalable architecture and modular approach were necessary to support future expansions, such as multi-campus deployment or extended functionality.

Action:

- Designed a modular architecture where each feature (e.g., User Management, Room Management) operates as an independent component.
- Integrated APIs, making the system adaptable for integration with external systems or apps.
- Utilized machine learning (Prophet) to predict room usage trends, which can guide decision-making beyond current applications.

Output: The system's scalability and extensibility allow it to be generalized for other domains like equipment reservations, event scheduling, or even corporate resource management. Its analytics and forecasting features support strategic planning and optimization of resources.

3. Purpose Alignment

Situation: The agreed purpose of the project was to develop a robust room booking system that simplifies user interactions while providing administrators with actionable insights to manage resources effectively.

Action:

- Delivered fully functional components aligned with user stories and functional requirements.
- Implemented role-based access control to provide a secure user experience.
- Developed an AI-powered chatbot for simplified navigation and quick responses to user queries.
- Enhanced administrative capabilities with features like exportable analytics and predictive modeling.

Output: The system aligns perfectly with its objectives by:

- Ensuring usability with responsive design and intuitive interfaces.
- Providing actionable insights through analytics and forecasting.
- Reducing administrative burdens through automation and data-driven decision-making.

Architecture Diagram

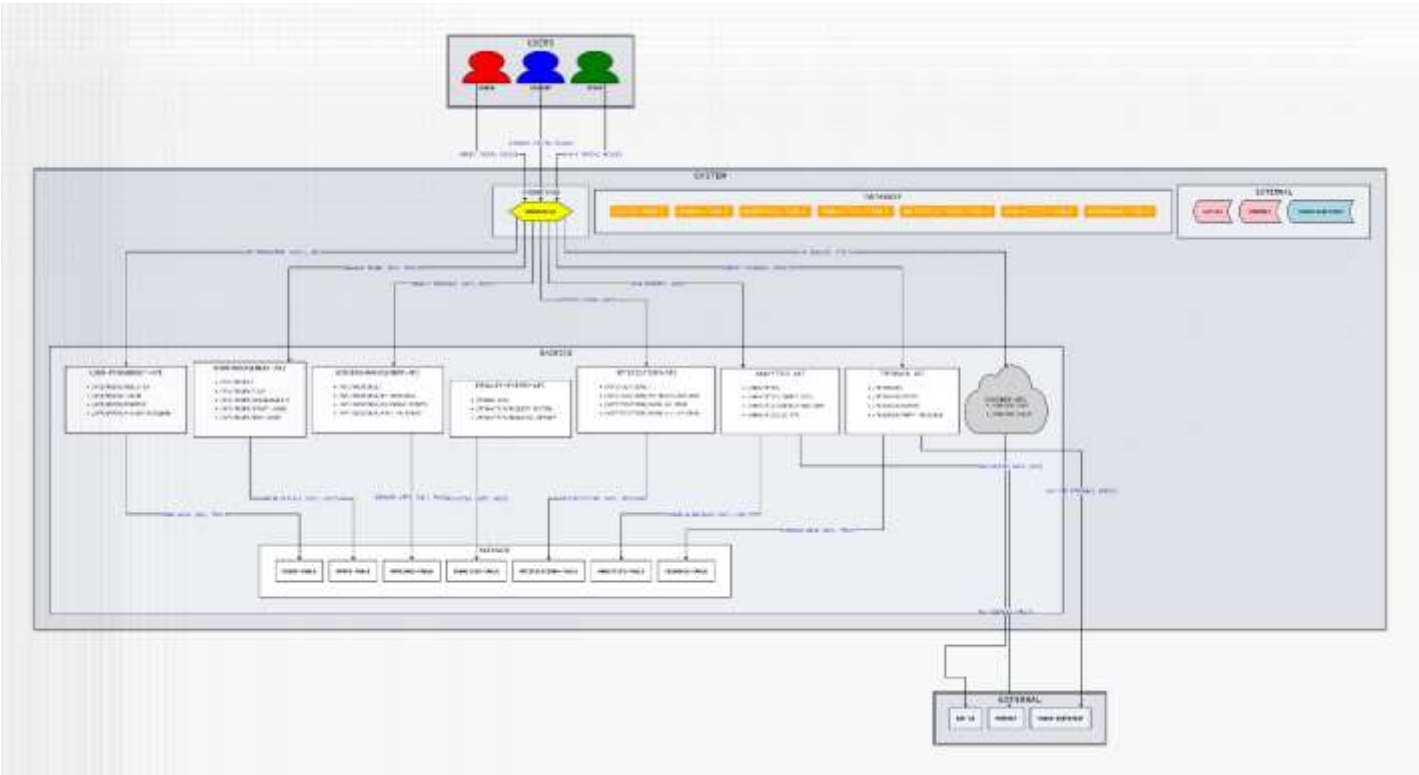


Figure 1

Risk Registry Table

Ris k ID	Risk Description	Category	Likeliho od	Impact	Risk Level	Mitigation Strategy	Contingency Plan
R1	Integration issues with Wit.ai for chatbot intent recognition.	Technical	Medium	High	High	Conduct API testing regularly; implement fallback responses when intent recognition fails.	Provide a static FAQ response in case Wit.ai services are unavailable.
R2	Data loss during database migration or schema updates.	Operational	Low	High	Mediu m	Back up the database before migration; test schema updates in a	Use backup data to restore the database if data is lost.

						staging environment before deploying to production.	
R3	Delayed implementation of the feedback system.	Project Management	Medium	Medium	Medium	Allocate dedicated team members to work on the feedback module; perform weekly progress reviews.	Deprioritize less critical features like customizable dashboards to focus on the feedback system.
R4	API rate limits imposed by Wit.ai or Prophet affecting functionality.	External	High	Medium	High	Implement API caching mechanisms to reduce the frequency of API calls; monitor API usage regularly.	Switch to alternative APIs or static responses temporarily until limits are reset.
R5	Unauthorized access to sensitive user data.	Technical	Low	High	Medium	Use JWT-based authentication; enforce strong password policies; perform periodic security audits.	Disable affected accounts immediately and notify users if a breach occurs.
R6	Server downtime affecting chatbot and analytics services.	Operational	Medium	High	High	Use a reliable hosting provider with an SLA; implement server monitoring	Switch to a backup server or deploy a temporary static version of critical features.

						and alert systems.	
R7	User dissatisfaction due to low sentiment accuracy in feedback.	Technical	Medium	Medium	Medium	Validate sentiment analysis using VADER with real-world feedback data; fine-tune thresholds for sentiment classification.	Provide manual sentiment review by admin for flagged cases.
R8	Miscommunication between team members delaying feature completion.	Project Management	Low	Medium	Medium	Conduct daily stand-up meetings; track progress using Monday.com or equivalent project management tools.	Escalate unresolved issues to the project lead for immediate resolution.
R9	Insufficient scalability for handling large or analytics data.	Technical	Medium	High	High	Use optimized database queries; implement pagination for APIs; ensure the database is properly indexed for high-performance queries.	Allocate additional server resources temporarily or migrate to a cloud service.
R10	Dependency on third-party tools (Wit.ai, Prophet) for critical features.	External	High	High	High	Develop modular APIs to allow switching to alternative services if needed; maintain a versioned	Replace third-party services with in-house algorithms or open-source alternatives in case of unavailability.

						backup of critical third-party dependencies.	
R11	Overstay penalties not being applied consistently.	Technical	Medium	Medium	Medium	Ensure robust testing of penalty logic; use automated test cases to validate different scenarios like no-shows and late cancellations.	Provide admin tools to manually impose penalties when inconsistencies are detected.
R12	Lack of real-time notifications for critical updates.	Technical	Medium	Medium	Medium	Use a reliable notification system with retry mechanisms; monitor for failed notifications and re-trigger them.	Notify users via alternative channels (e.g., email) if real-time notifications fail.
R13	Difficulty understanding chatbot responses.	Usability	Medium	Medium	Medium	Perform user testing to validate chatbot conversations; train the chatbot to handle multiple phrasing variations of the same question.	Provide a fallback response with links to detailed documentation or FAQ.
R14	Legal regulatory compliance or for	Legal	Low	High	Medium	Ensure the system adheres to GDPR or similar	Seek legal counsel if compliance issues arise; revise terms

	user data handling.					regulations for user data; provide users with clear data collection and retention policies.	and conditions as necessary.
R15	User confusion with role-based booking restrictions.	Usability	Low	Medium	Low	Provide clear instructions on role-based restrictions during registration and booking processes; show error messages with explanations.	Allow admins to override restrictions in exceptional cases.

Table 4

User Management

Test Case ID	Description	Steps	Expected Outcome	Actual Outcome	Result
UM001	User Registration	Register with valid email, username, password, and role.	User is successfully registered.	User successfully registered.	Pass
UM002	Duplicate Registration	Register with an already existing email.	Error message: "Email already registered."	Error message displayed as expected.	Pass
UM003	User Login	Login with valid credentials.	User receives a JWT access token and role information.	JWT token and role information received.	Pass
UM004	Invalid Login	Attempt to login with incorrect password.	Error message: "Invalid email or password."	Error message displayed correctly.	Pass
UM005	Profile Retrieval	Retrieve profile details using email.	User details are retrieved successfully.	User details retrieved successfully.	Pass

Table 5

Room Management

Test Case ID	Description	Steps	Expected Outcome	Actual Outcome	Result
UM001	User Registration	Register with valid email, username, password, and role.	User is successfully registered.	User successfully registered.	Pass
UM002	Duplicate Registration	Register with an already existing email.	Error message: "Email already registered."	Error message displayed as expected.	Pass
UM003	User Login	Login with valid credentials.	User receives a JWT access token and role information.	JWT token and role information received.	Pass
UM004	Invalid Login	Attempt to login with incorrect password.	Error message: "Invalid email or password."	Error message displayed correctly.	Pass
UM005	Profile Retrieval	Retrieve profile details using email.	User details are retrieved successfully.	User details retrieved successfully.	Pass

Table 6

Penalty System

Test Case ID	Description	Steps	Expected Outcome	Actual Outcome	Result
PS001	Penalty Application	Apply a penalty for a no-show booking.	Penalty is created and linked to the user's account.	Penalty was successfully created and linked.	Pass
PS002	Penalty Resolution	Admin resolves a penalty for a user.	Penalty status changes to "resolved."	Penalty status updated to "resolved."	Pass
PS003	Penalty Reporting	Generate a penalty report in CSV format.	CSV file contains correct penalty details.	CSV file generated with accurate details.	Pass

Table 7

Analytics & Reporting

Test Case ID	Description	Steps	Expected Outcome	Actual Outcome	Result
AR001	Room Usage Trends	Fetch usage data for a specific time range.	Usage trends are aggregated and returned.	Usage trends successfully aggregated and returned.	Pass
AR002	Export Analytics Report	Export analytics data to CSV.	CSV file is downloaded with correct details.	CSV file downloaded with accurate data.	Pass
AR003	Forecast Room Usage	Generate room usage forecast for the next 30 days.	Forecasted usage data is returned successfully.	Room usage forecast generated successfully.	Pass

Table 8

Chatbot

Test Case ID	Description	Steps	Expected Outcome	Actual Outcome	Result
AR001	Room Usage Trends	Fetch usage data for a specific time range.	Usage trends are aggregated and returned.	Usage trends successfully aggregated and returned.	Pass
AR002	Export Analytics Report	Export analytics data to CSV.	CSV file is downloaded with correct details.	CSV file downloaded with accurate data.	Pass
AR003	Forecast Room Usage	Generate room usage forecast for the next 30 days.	Forecasted usage data is returned successfully.	Room usage forecast generated successfully.	Pass

Table 9

Feedback

Test Case ID	Description	Steps	Expected Outcome	Actual Outcome	Result
AR001	Room Usage Trends	Fetch usage data for a specific time range.	Usage trends are aggregated and returned.	Usage trends successfully aggregated and returned.	Pass
AR002	Export Analytics Report	Export analytics data to CSV.	CSV file is downloaded with correct details.	CSV file downloaded with accurate data.	Pass

AR003	Forecast Room Usage	Generate room usage forecast for the next 30 days.	Forecasted usage data is returned successfully.	Room usage forecast generated successfully.	Pass
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Table 10

Risk Mitigation Analysis

1. **Integration issues with Wit.ai:** The API testing schedule was followed diligently and we implemented automated fallback responses for when train disruption occurred which allowed the system to work flawlessly.
2. **Data loss during database migration:** The team created backups ahead of schema modifications while staging-testing all updates in a separate environment before moving them to production.
3. **User dissatisfaction with chatbot accuracy:** We performed user testing which enabled us to refine responses with actual user queries to enhance the chatbot execution speed.
4. **Dependency on third-party tools:** To enable fluid transitions during system failure modular APIs operated as a failover system that facilitated rapid replacement with alternative services.

Team Collaboration

Scenario

A team of five final-year students is developing a web application for an **Intelligent Spatial Management System**, specifically a **Library Room Booking System**. The system allows users to book rooms and provides administrators with the ability to manage room details (add, remove, update rooms). A standout feature of the system is an **AI-powered chatbot**, which assists users with queries and bookings.

The project is to be completed within **10 weeks**, aligned with strict academic term timelines and assessment criteria.

Team Composition

1. Team Leader (Project Coordinator)

- Manages project timelines and tracks progress.
- Handles both backend and frontend development.
- Ensures alignment with project goals.

2. UI/UX Designer

- Designs the system's interface and enhances user experience.
- Focuses on clarity and ease of use.

3. Backend Developer

- Develops server-side logic and manages database integration.
- Implements AI chatbot functionality.

4. Frontend Developer

- Implements UI/UX designs and integrates frontend functionalities.
- Ensures responsiveness and usability.

5. Quality Assurance (QA) Lead

- Ensures the system is bug-free and meets user requirements.
- Conducts testing and validation.

Communication and collaboration tools

- **WhatsApp** – Used for daily team communication and quick updates.
- **Microsoft Teams** – Used for meetings, calls, and detailed project discussions.
- **Clickup**– Recommended for task tracking and documentation.

WhatsApp

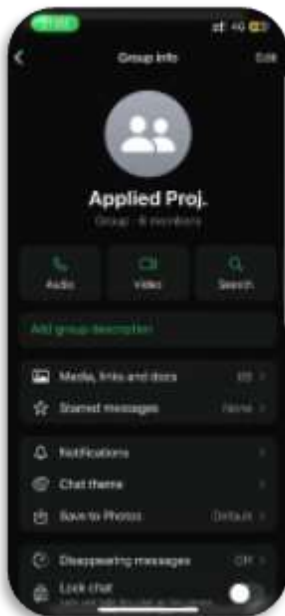


Figure 2

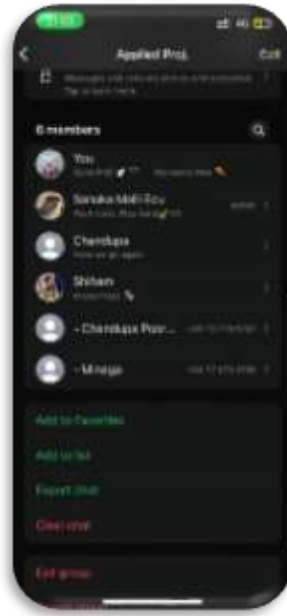


Figure 3



Figure 4

Clickup

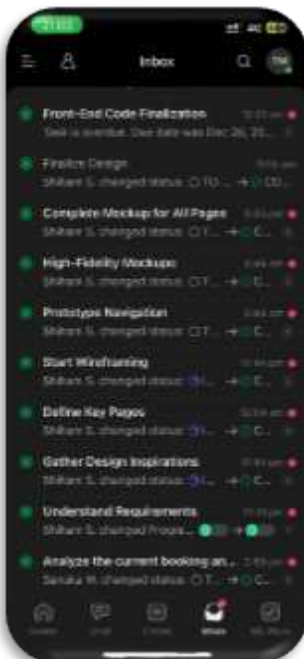


Figure 5



Figure 6

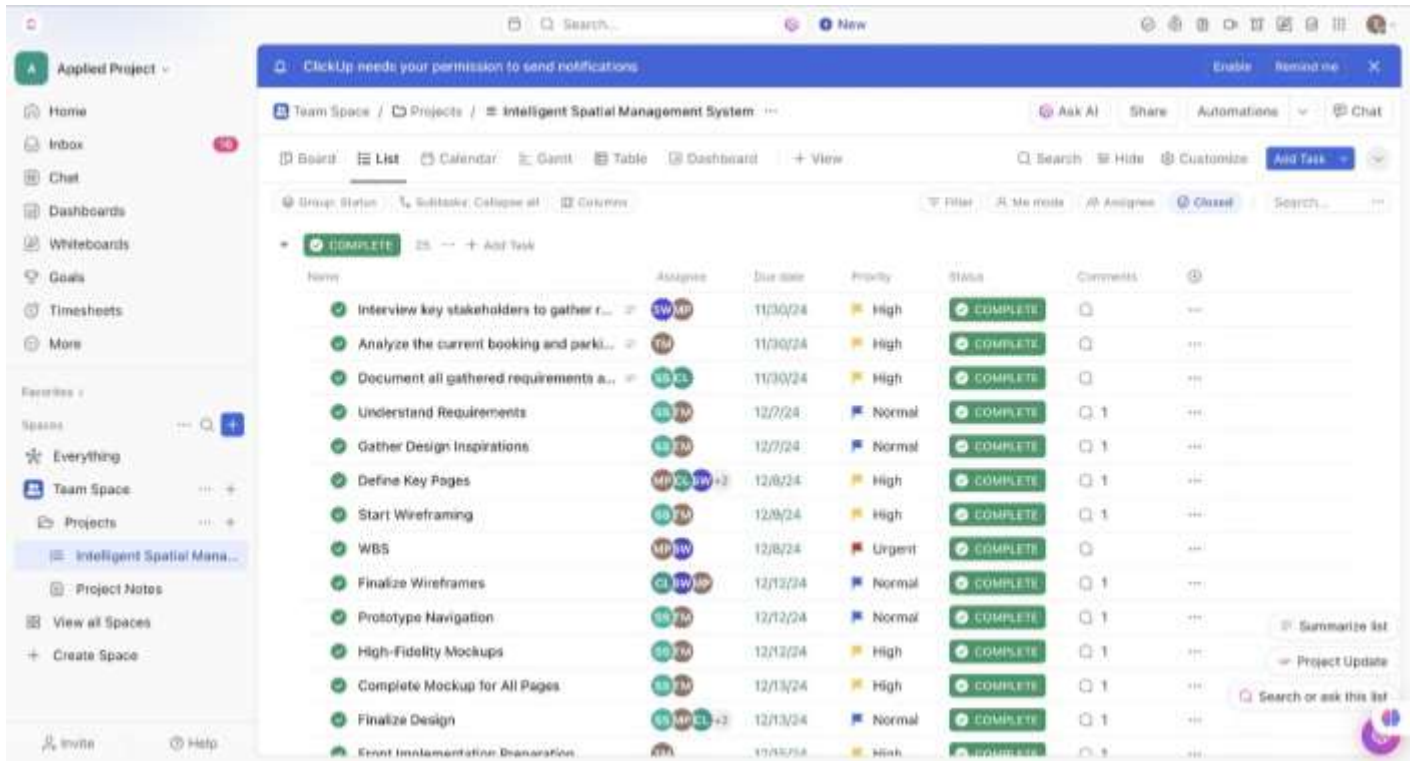


Figure 7

Teams



Figure 8

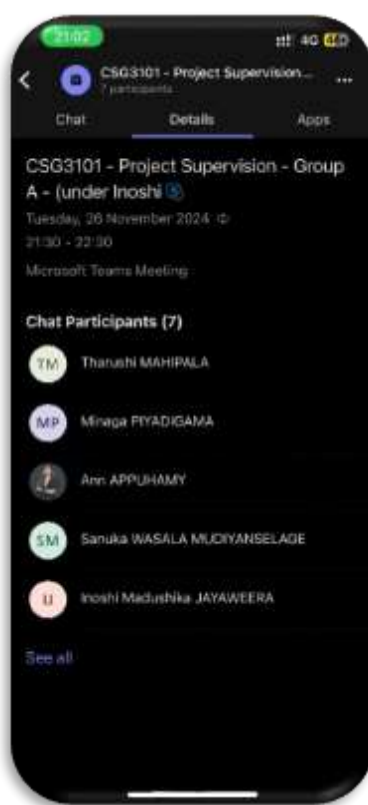


Figure 9

Meeting Details

Main Meetings that took with the Supervisor.

Meeting 1

Date	Nov 28, 2023
Time	3:00 PM
Location	Teams
Present	Sanuka, Minaga, Chandupa, Shihaam, Umesha
Absent	None
Discussion	- Initial project planning - Task allocation
Decisions	- Agreed on roles - Finalized tools
Actions	- Create task breakdown - Assign tasks to team members
Next Meeting	Dec 5, 2023, Teams

Table 11

Meeting 2

Date	Dec 5, 2023
Time	3:00 PM
Location	Teams
Present	Sanuka, Minaga, Chandupa, Shihaam, Umesha
Absent	None
Discussion	- Review of task progress - Discuss challenges
Decisions	- Revised timelines - Addressed blockers
Actions	- Complete initial modules - Report progress
Next Meeting	Dec 12, 2023, Teams

Table 12

Meeting 3

Date	Dec 12, 2023
Time	3:00 PM
Location	Teams
Present	Sanuka, Minaga, Chandupa, Shihaam, Umesha
Absent	None
Discussion	- Feedback on module development - Planning next sprint
Decisions	- Accepted modules - Set deliverables
Actions	- Implement feedback - Prepare for user testing

Next Meeting	Dec 19, 2023, Teams
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Table 13

Meeting 4

Date	Dec 19, 2023
Time	3:00 PM
Location	Teams
Present	Sanuka, Minaga, Chandupa, Shihaam, Umesha
Absent	None
Discussion	- User testing feedback - Debugging issues
Decisions	- Agreed on fixes - Updated project plan
Actions	- Address bugs - Prepare demo videos
Next Meeting	Jan 2, 2024, Teams

Table 14

Meeting 5

Date	Jan 2, 2025
Time	3:00 PM
Location	Teams
Present	Sanuka, Minaga, Chandupa
Absent	Shihaam, Umesha
Discussion	- New Year update - Remaining modules discussion
Decisions	- Finalized pending tasks - Realigned timeline
Actions	- Finalize feedback module - Conduct integration testing
Next Meeting	Jan 9, 2025, Teams

Table 15

Meeting 6

Date	Jan 9, 2025
Time	3:00 PM
Location	Teams
Present	Sanuka, Minaga, Chandupa, Shihaam, Umesha
Absent	None
Discussion	- Integration progress - Deployment preparation
Decisions	- Finalized deployment plan - Scheduled presentations
Actions	- Ensure complete testing - Prepare final report
Next Meeting	Jan 16, 2025, Teams

Table 16

Meeting 7

Date	Jan 16, 2025
Time	3:00 PM
Location	Teams
Present	Sanuka, Minaga, Chandupa, Shihaam, Umesha
Absent	None
Discussion	- Presentation preparation - Final report review
Decisions	- Agreed on demo structure - Assigned presentation parts
Actions	- Record final demos - Submit report
Next Meeting	Jan 23, 2025, Teams

Table 17

Physical Meetings with Team Members

Meeting 01

Meeting 02

Date	Dec 06, 2024
Time	9.00AM
Location	On campus
Present	Sanuka, Minaga, Chandupa, Shihaam
Absent	Umesha
Discussion	-Discuss about project process - Discuss challenges
Decisions	-Finalize project proposal
Actions	-Start front end coding
Next Meeting	Dec 18, 2024

Table 19

Meeting 03

Date	Dec 18, 2024
Time	1.00PM
Location	On campus
Present	Sanuka, Minaga, Chandupa, Shihaam, Umesha
Absent	None
Discussion	-Backend Progression -Finalize UI -Frontend Progression
Decisions	-Start Backend development
Actions	- Start Backend coding - Continue frontend coding
Next Meeting	Dec 22, 2024

Table 20

Meeting 04

Date	Dec 22, 2024
Time	1.00PM
Location	On campus
Present	Sanuka, Minaga, Chandupa, Shihaam, Umesha
Absent	None
Discussion	-Discuss on chatbot development -Discuss on Frontend and backend development
Decisions	-Finalize chatbot development plan
Actions	-Start Chatbot development -Continue Backend and Frontend Development
Next Meeting	Dec 28, 2024 , On campus

Table 21

Meeting 05

Date	Dec 28, 2024
Time	1.00PM
Location	On campus
Present	Sanuka, Minaga, Chandupa, Shihaam, Umesha
Absent	None
Discussion	-Discuss about development progression
Decisions	-Finish frontend development within first week of jan
Actions	-Start Bug fixes and final development of frontend -Continue backend and almost finish
Next Meeting	Jan 04, 2025, On campus

Table 22

Meeting 06

Date	Jan 04, 2025
Time	1.00PM
Location	On campus
Present	Sanuka, Minaga, Chandupa, Shihaam
Absent	Umesha
Discussion	-Finalize frontend -Finalize chatbot
Decisions	-Do test run of the frontend and chatbot
Actions	-Fix bugs in frontend and chatbot -Start Make frontend responsive
Next Meeting	Jan 10, 2025 On campus

Table 23

Meeting 07

Date	Jan 10, 2025
Time	9.00AM
Location	On campus
Present	Sanuka, Minaga, Chandupa, Shihaam, Umesha
Absent	None
Discussion	-Continue Testing -Discuss on Project report
Decisions	-Continue Testing and bug fixing start doing report
Actions	-Start Project report
Next Meeting	Jan 18,2025

Table 24

Meeting 08

Date	Jan 18, 2025
Time	9.00AM
Location	On campus
Present	Sanuka, Minaga, Chandupa, Shihaam, Umesha
Absent	None
Discussion	-Discuss remaining bugs and do a test run again
Decisions	-Continue Report and also Bug fixing
Actions	-95% of project almost complete
Next Meeting	Jan 22,2025

Table 25

Meeting 09

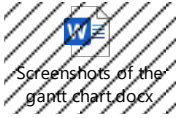
Date	Jan 22, 2025
Time	9.00AM
Location	On campus
Present	Sanuka, Minaga, Chandupa, Shihaam, Umesha
Absent	None
Discussion	-Discuss about the presentation
Decisions	-Presentation preparation - Final report review
Actions	- Record final demos - Submit report
Next Meeting	Jan 25,2025

Table 26

Gantt Chart

Initial Gantt Chart

Below are the screenshots of the Initial Gantt Chart



Link for the Gantt chart screenshots

https://drive.google.com/drive/folders/1AMoAeub41kshEIJYmF534MsZlGFpQwSq?usp=drive_link

Report on UI/UX Development and Timeline Adjustments

During the initial phase of the UI/UX development, there was a lack of consensus among the group members regarding the design direction. This disagreement led to a conflict about whether to proceed with the existing design or to redesign the entire frontend. A key factor in this decision was the potential impact on the project's timeline, particularly the Gantt chart.

The primary concern was that redesigning the entire frontend could disrupt the established timeline. However, after thorough discussions and consideration, the group reached a collective decision. We determined that the frontend of the application is the first interaction users will have with the platform, and therefore, it would significantly influence the user's first impression. To ensure a seamless and engaging user experience, the frontend development was prioritized over sticking to the original timeline.

As a result of this decision, the initial Gantt chart timeline was disrupted and slightly rearranged. Initially, there were only two frontend developers assigned to the task. However, recognizing the importance of frontend design and the need for effective resource distribution, it was decided to involve all five group members in the frontend development process. This shift in workload distribution was a major reason behind the changes made to the Gantt chart.

The final Gantt chart, reflecting these updates and adjustments, is provided below, with the tasks redistributed across all five team members.

Actual Gantt Chart

Intelligent Spatial Management System
Group 08 Project lead

Project start: Thu, 11/28/2024
Display week: 1

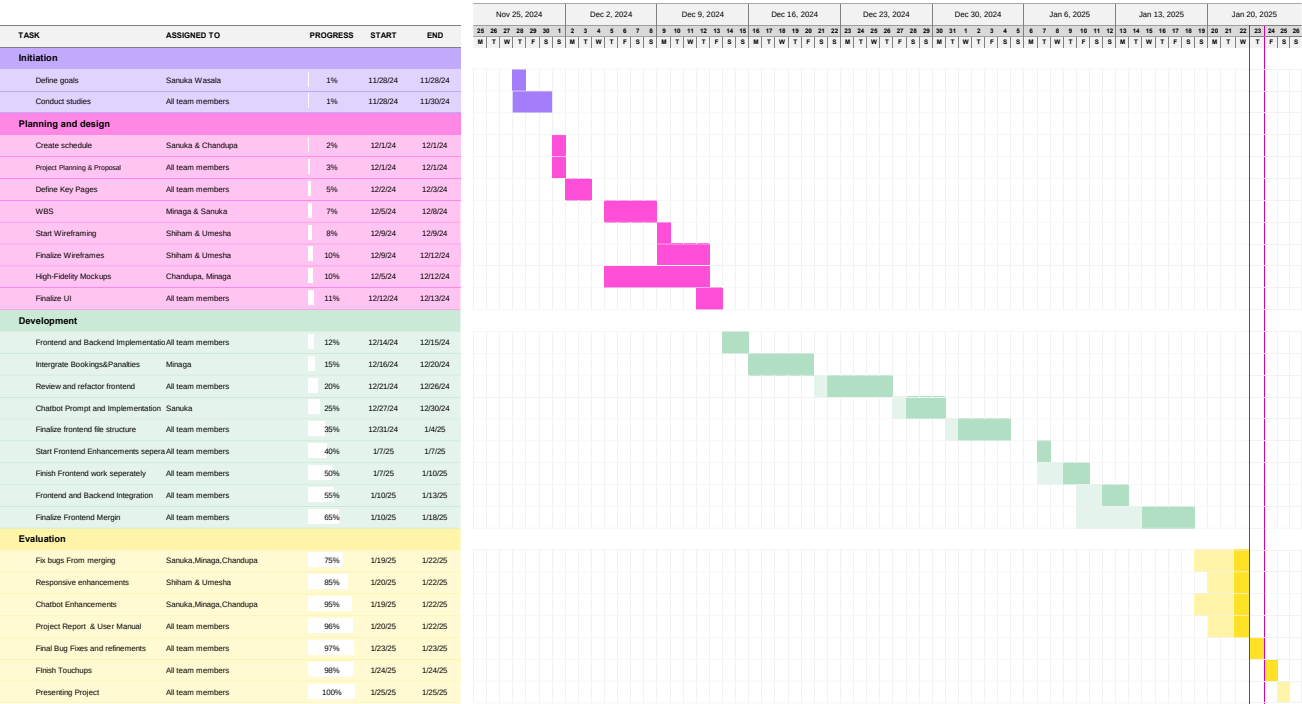


Figure 9

Summary for the Gantt chart

Executive Summary

The team at Applied Project has successfully completed a series of tasks related to the Intelligent Spatial Management System, including gathering design inspirations, defining key pages, and conducting stakeholder interviews to gather requirements. They have also completed tasks such as prototyping navigation, wireframing, and creating high-fidelity mockups, ensuring a comprehensive approach to the project development.

All projects appear to be on track, with no current indicators of risk or delay. The tasks have been completed within their respective timelines, suggesting effective project management and execution.

Key Efforts & Initiatives

- **Design and Prototyping:** Developing the visual and functional design of the project.
 - Gather Design Inspirations
 - Define Key Pages
 - Prototype Navigation
 - Complete Mockup for All Pages
 - Finalize Design
- **Requirement Gathering and Analysis:** Collecting and analyzing requirements to inform project development.
 - Interview key stakeholders to gather requirements (students, staff, admins).
 - Document all gathered requirements and summarize findings for backlog creation
 - Understand Requirements
 - Analyze the current booking and parking processes to identify inefficiencies
- **Front-End Development:** Preparing and finalizing the front-end implementation of the project.
 - Front Implementation Preparation
 - Review & Refactor Frontend Code
 - Front-End Code Finalization

Other tasks include wireframing, creating high-fidelity mockups, and integrating booking and penalties logic.

Resource Management

Task Allocation and Team Responsibilities

Our project distributed roles and responsibilities amongst team members to facilitate efficient system development and proper delivery of the Intelligent Spatial Management System. The allocation of work responsibilities followed both member capabilities and project necessities.

1. Team Leadership

Sanuka (Tech Lead, Full stack developer)

- **Situation:** Team leadership exists to reach project goals by following schedule deadlines and technical requirements at maximum quality and problem resolution standards.
- **Action:** The leader managed his team while coordinating group meetings and tracking progress together with overseeing backend development work tasks.
- **Output:** The project manager worked to reach all scheduled deadlines alongside resolving problems right away and kept all functions united.

2. Backend Development

Sanuka, Minaga, Chandupa

- Situation: The system needed substantial backend engineering to execute booking operations while controlling user access controls and generate analytics.
- Action: The team implemented an API architecture and database functionality using Python Django to create the backend for the system logic and enhanced connectivity between platform elements.
- Output: The newly developed backend system provides both an efficient platform for booking room spaces with secure ways to manage all user actions.

3. Frontend Development & UI/UX Design

Umesha, Shiham

- Situation: The system required an intuitive user interface design to deliver an effortless user experience for those who used it.
- Action: The team created the frontend components through HTML and CSS and JavaScript while adopting the minimalistic principles of Apple design.
- Output: The system presents an intuitive front-end application with good visual appeal for room booking operations.

4. Quality Assurance (QA)

Minaga, Chandupa

- Situation: Testing of the system proceeded with recommended methods to confirm performance levels and system reliability.
- Action: The team executed extensive system evaluation which comprised functional checks along with performance measurement and usability analysis.
- Output: The team evaluates and fixes system faults during quality assurance to improve interface performance and system usability.

Hardware Components

The development and testing of the Intelligent Spatial Management System needed multiple hardware resources to guarantee operational efficiency and system responsiveness as well as overall reliability. The chosen hardware resources matched the requirements for supporting development operations while enabling teamwork activities and providing extensive testing capability throughout different platforms.

1. Hardware Resources Used

- **Laptops:**
All team members accessed development tasks through Windows laptops equipped with Intel i5 and higher processors so they could properly code and operate local servers and databases.
- **Campus Study Room TV:**
The team utilized the campus study room television to demonstrate and test their web application as they checked visual display quality and ease of use for real-world conditions.
- **Mobile Phones:**
Various smartphones were utilized for screening the application responsiveness across different screen dimensions to confirm compatibility with smaller devices.
- **Tablets:**
Our team tested system performance alongside screen layout through tablet devices to maintain peak usability throughout different device sizes.

2. Justification for Chosen Hardware

- **Performance and Efficiency:**
Laptop laptops with Intel i5 processors as minimum standard delivered a powerful computing base running development tools including Django and MySQL along with Figma design software in a performant environment.
- **Scalability and Testing:**
The study room TV provided staff members with a large screen display to present and monitor the application which facilitated design fault detection and improved overall user interface and experience.
- **Cross-Device Responsiveness:**
The choice to use mobile devices and tablets during development resulted in applications rendering properly across various screen dimensions thus creating enhanced user accessibility and usability.

Software components

Several tools and technologies supported the development of the Intelligent Spatial Management System and worked together to establish a clear workflow structure for both development and communication. We based our selection on these tools to enable collaboration and simplify development work.

1.Tools & Technologies Used

Technology	Purpose
Figma	UI/UX design and prototyping
Django	Backend development and API implementation.
HTML, CSS, JavaScript	Frontend Development
MySQL Workbench	Database management and queries.
Microsoft Teams	Microsoft Teams
WhatsApp	Daily Communications and Quick updates.
Click up	Task Tracking & Project Management
Postman	API testing and Debugging

Table 27

2. Importance and Contribution to Project Success

Each software tool played a critical role in the successful completion of the project:

- HTML, CSS, and JavaScript:
 - The core frontend technologies combined to deliver responsive interfaces alongside interactive elements and natural user interactions.
 - Using these software tools designers developed a simplified user interface which followed Apple's distinct design principles.
 - CSS managed the styling component of design though it also provided consistent display across devices and JavaScript took charge of dynamic elements including API integrations.
- Figma:
 - Through Figma our team worked together to improve and design the user interface so users experienced a reliable and easy-to-use platform.
 - We used the design visualization feature to eliminate development mistakes prior to implementation.
- Django:
 - This framework supplied a fundamental development structure enabling project scale and essential security capabilities.
 - The system supported straightforward API development and unified the backend elements with the frontend requirements.
- MySQL Workbench:
 - The platform enabled productive database schema and queries management functionality.
 - Designers achieved two functions using the system: they could simplify data models and gain performance understanding about databases.
- Microsoft Teams:
 - Microsoft Teams enabled effective virtual collaboration through structured meetings with additional file sharing capabilities.

- Clear project discussions received documented support for proper communication both through documentation.
- WhatsApp:
 - The rapid and unstructured messaging system allowed us to exchange urgent messages and organize teams quickly.
 - Our team stayed on the same page with daily operations through successful mutual communication.
- Click Up:
 - The platform enabled us to track tasks and monitor project progress and control deadlines.
 - This tool efficiently managed project visibility by tracking both statuses and assigned tasks.
- Postman:
 - My work involved testing the API while validating backend features to fulfill project specifications through Postman.
 - API response testing determined and corrected bugs before major development commenced.

3. Challenges Faced and Their Resolutions

During the project, the team encountered several challenges related to the usage of software tools:

- Challenge:
- The project needed to deliver consistent user interfaces that operated flawlessly on multiple devices along with various browsers.
 - Resolution:
 - A comprehensive testing process using HTML CSS and JavaScript adjustments raised the system's compatibility levels.
 - The team used media queries together with responsive design techniques to boost the system's adaptive features.
- Challenge:
- Team members with differing devices that reside in various time zones needed to synchronize both interface redesigns and backend modifications.
 - Resolution:
 - Real-time update tracking used Figma's collaborative features.
 - Our team holds structured review conferences on Teams each week to maintain a coordinated approach.
- Challenge:
- Teams must effectively manage and set precedence for their tasks during multiple development phases.
 - Resolution:
 - A ClickUp implementation allowed our team to set rank order for tasks alongside deadline assignments.
 - We used daily stand-up discussions through WhatsApp as a method to maintain project alignment.
- Challenge:

- The team needed to verify API integration functionality with absolute consistency between various execution environments.
 - Resolution:
 - Used Postman for consistent testing across different systems.
 - The backend was developed to implement RESTful best practices throughout its design.

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